

Name : Sk Arshad Basha

Reg no : 192312336

Implement Perceptron based IRIS classification

Code :

```
# Perceptron IRIS Classification
```

```
X = [[5, 3], [4, 2], [6, 3], [7, 3]]
```

```
Y = [0, 0, 1, 1]
```

```
w1 = 0
```

```
w2 = 0
```

```
b = 0
```

```
lr = 0.1
```

```
for epoch in range(10):
```

```
    for i in range(len(X)):
```

```
        y = w1*X[i][0] + w2*X[i][1] + b
```

```
        pred = 1 if y >= 0 else 0
```

```
        error = Y[i] - pred
```

```
        w1 += lr * error * X[i][0]
```

```
        w2 += lr * error * X[i][1]
```

```
        b += lr * error
```

```
test = [6, 3]
```

```
y = w1*test[0] + w2*test[1] + b
```

```
pred = 1 if y >= 0 else 0
```

```
print("Predicted Class:", pred)
```

```
if pred == 0:
```

```
    print("Flower: Setosa")
```

```
else:
```

```
    print("Flower: Versicolor")
```

Output :

```
Predicted Class: 1
```

```
Flower: Versicolor
```

```
=== Code Execution Successful ===
```